Welding Processes

1. **Welding** is a fabrication process that joins materials, usually metals or thermoplastics, by causing coalescence. (to grow together or into one body)

2. This is often done by melting the work pieces and adding a filler material to form a pool of molten material (**the weld pool**) that cools to become a strong joint, with pressure sometimes used in conjunction with heat, or by itself, to produce the weld.

3. Welding is used in every industry large or small, as a principle means of fabrication and repairing metal product.

4. The process is efficient, economical and dependable.

5. This is the only process which has been tried in the space and finds applications for fabricating products in air, underwater and in space

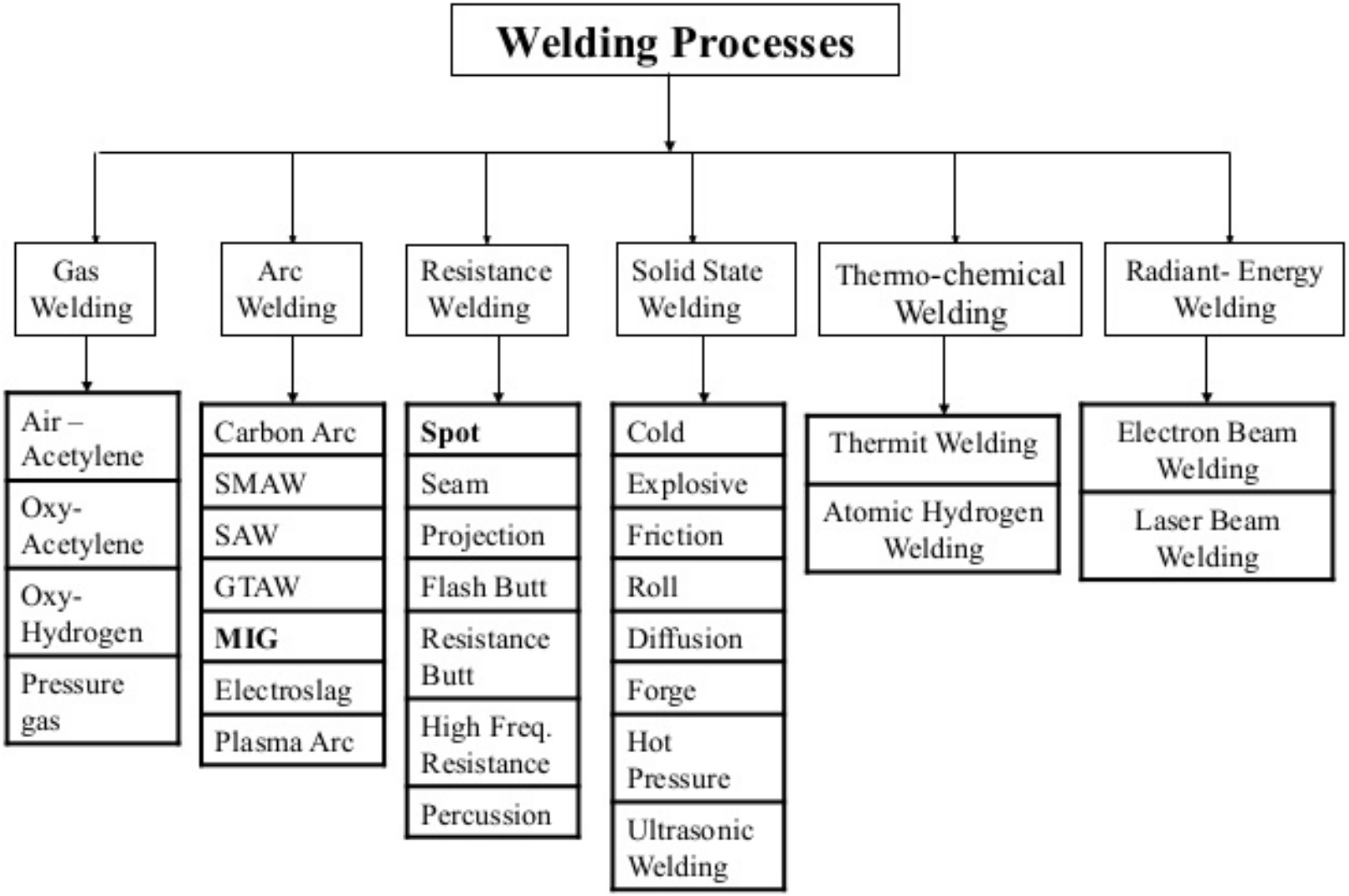
Welding

- In this metal joining process the two parts that are to be welded are fused together by application of heat and pressure (sometimes).
- Permanent Fusion happens between joining metals
- The welding processes depends on many factors depending on the type of welding process:
  - Oxy-acetylene welding
  - SMAW
  - GMAW
  - GTAW
  - FCAW; SAW; ESW; ERW etc….

Welding is a process of permanently joining materials. Welding joins different metals/alloys with a number of processes, in which heat is supplied either electrically or by means of a torch. Welding is done by application of heat or both heat and pressure. The most essential requirements is Heat. Pressure may be employed, but this is not in many processes essential.

The welding process evolves applying heat to the work piece. The heat applied should be such that the work piece should melt, i.e. the temperature at which welding is done, should be more than the melting point of the work piece to be welded.

Classification of Welding Processes



Classification of Welding Process

A simple and logical way is to satisfy the welding processes according to the sources of energy used for achieving coalescence.

The welding processes can thus be broadly classified as;

(a) Fusion welding
(b) Pressure welding
(c) Thermo chemical welding

(a) Fusion welding

(1) Gas flame welding

(a) Oxy-acetylene welding
(b) Oxy-hydrogen welding
(c) Oxy-acetylene welding
(d) Pressure Gas welding

2. Electric arc welding

(a) Carbon arc welding

(b) Shielded metal arc welding

(c) Submerged arc welding

(d) Tungsten arc welding (TIG)

(e) Metal inert gas (MIG) welding

(f) Plasma arc welding

3. Radiant energy welding

(a) Electron beam welding

(b) Laser beam welding

4. Electric resistance welding

(a) Spot welding
(b) Seam welding
(c) Projection welding
(d) Resistance butt welding
(e) Flash welding
(f) Percussion welding
(g) Electroslag welding

Solid state (cold) welding

(b) Pressure welding

(a) Friction welding
(b) Ultrasonic welding
(c) Explosive welding
(d) Forge and diffusive welding

(c) Thermo chemical welding

(a) Thermit welding
(b) Atomic hydrogen welding

Fusion welding

Fusion welding is a group of processes that bond metals together by heating a portion of each piece above the melting point and causing them to flow and fuse together. Usually filler materials are used to insure the joint is filled. All fusion welding processes have four requirements for obtaining satisfactory welds.

1. A source of energy to create union (e.g. gas flame, electric arc, electric resistance etc.)
2. Removing surface contaminants (e.g. organic and oxide films etc.)
3. Protecting metal from atmospheric contamination (shielding gas atmosphere like argon, helium, carbon dioxide, smoke of burning flux coating)
4. Control of weld metallurgy (e.g. preheat, post welding heat treatment)

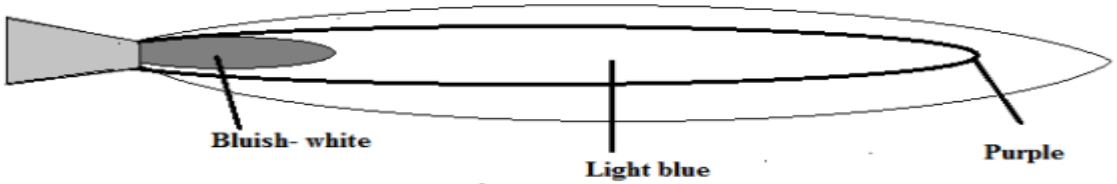
Oxy-acetylene fusion welding

Types of flame

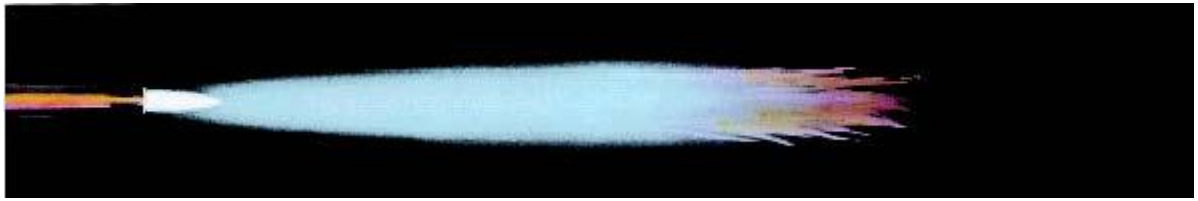
To achieve successful and efficient welding, proper mixing of gas in appropriate proportion is necessary. According to the volume of O_2 and C_2H_2 flame can be classified as

Figure - Gas welding flame

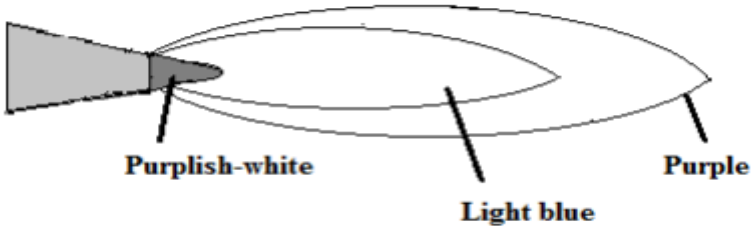
Neutral flame- Neutral Flame is produced when equal volume of O_2 and C_2H_2 are mixed. It has well defined inner cone, which has light blue color. Neutral Flame doesn't create any change in the molten metal and doesn't oxidize or carburize the metal. This flame is used for welding of mild steel, cast iron, aluminum and stainless steel. Temperature of neutral flame is 3232ºC.



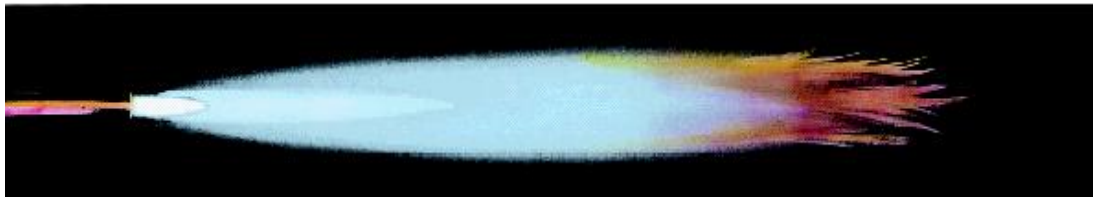
(a) Neutral flame (3232^0 C)



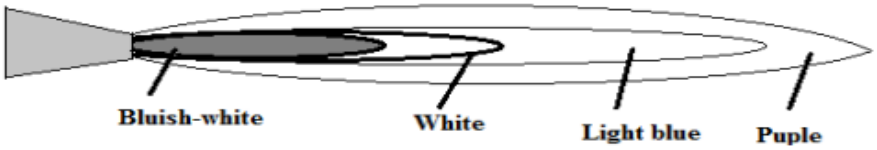
Oxidizing flame- oxidizing flame is obtained by increasing the supply of oxygen in mixture. It has small white cone which is shorter and more pointed than that of neutral flame. It creates high temperature of 3482ºC. It is normally not used except in the case of brass.



(b) Oxidizing flame (3482^0 C)



Reducing or Carburizing flame- It is produced when supply of C_2H_2 is higher than supply of oxygen and can create an approximate temperature of 3150ºC. its outer envelope is longer than that of neutral flame. It is used for the welding of lead. In these flames the temperature is maximum at the tip of the inner cone. Feather or brush exists next to the cone and occurs in carburizing flame only.

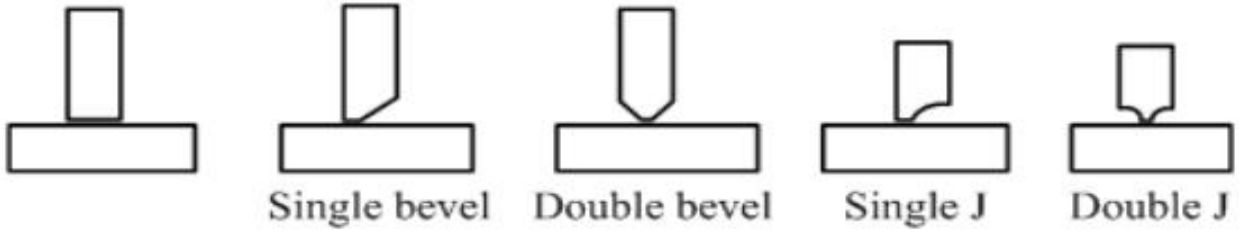


(c) Reducing flame(3150^0 C)

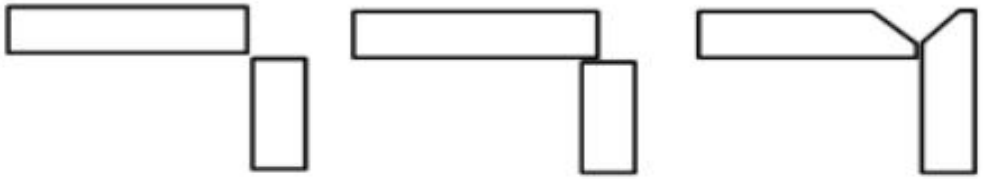
Weld joints

Butt welds

Tee welds

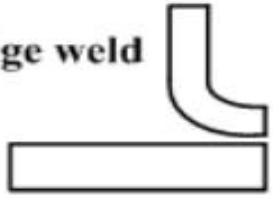


Corner welds



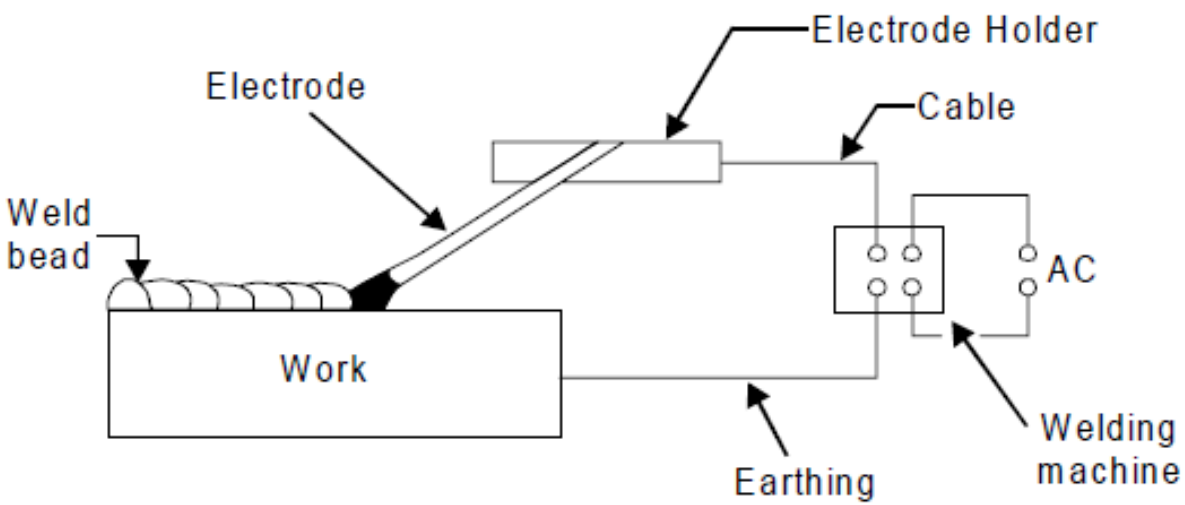
Lap weld

Edge weld



Arc Welding

An arc is generated between two conductors of electricity, cathode and anode (considering direct current, DC), **when they are touched to establish the flow of current and then separated by a small distance**. ... An arc is a sustained electric discharge through this ionised gas column called plasma between the two electrodes.



4. Welding Electrodes

An electrode is a piece of wire or a rod of a metal or alloy, with or without coatings. An arc is set up between electrode and workpiece. Welding electrodes are classified into the following types

1. Consumable Electrodes

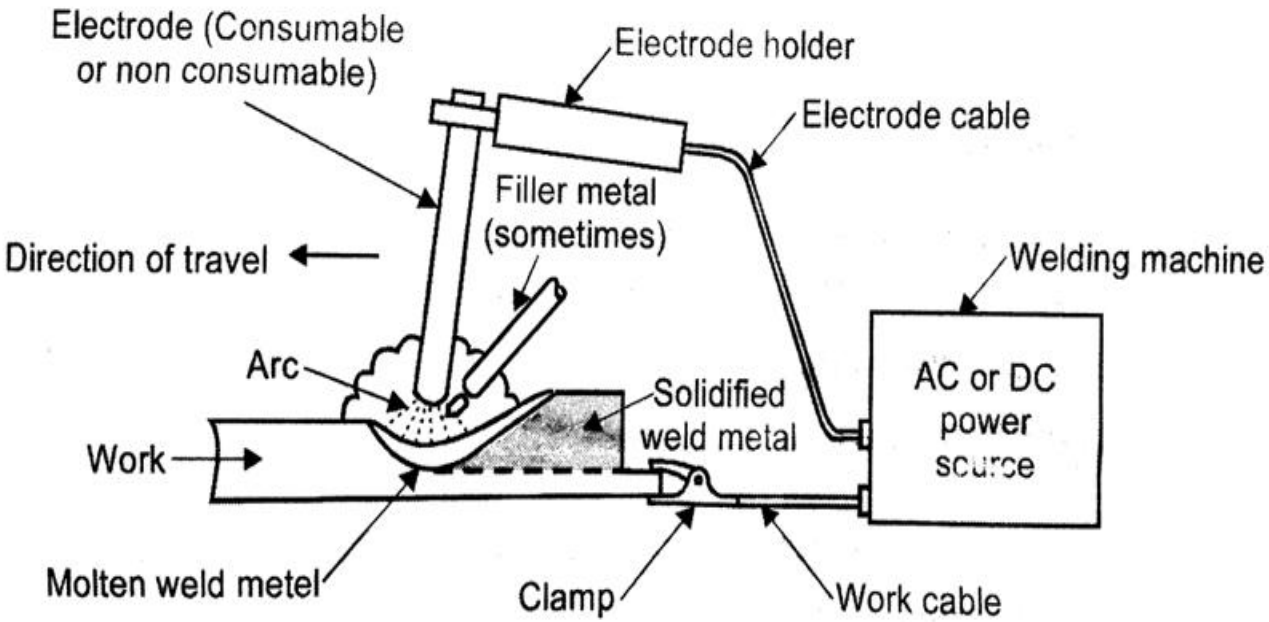
- Bare Electrodes
- Coated Electrodes

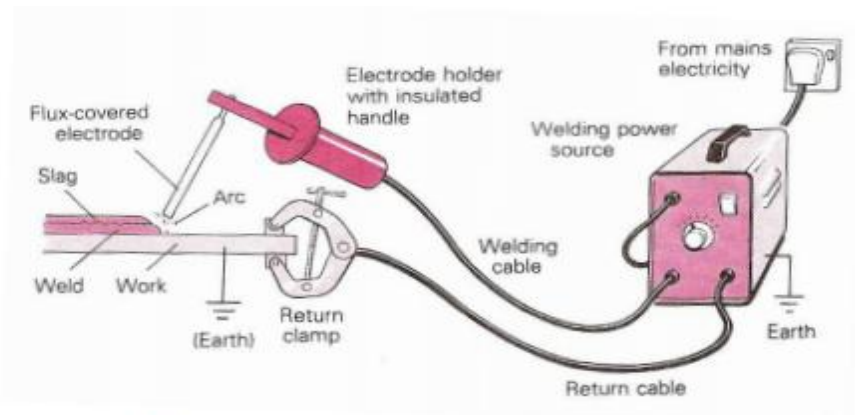
2. Non-consumable Electrodes

- Carbon or Graphite Electrodes
- Tungsten Electrodes

Consumable electrodes are made up of different metals and their alloys. The end of this electrode starts melting when the arc is struck between the electrode and workpiece. Thus consumable electrode itself acts as a filler metal. Bare electrodes consist of a metal or alloy wire without any flux coating on them. Coated electrodes have flux coating which starts melting as soon as an electric arc is struck. This coating on melting performs many functions like prevention of joint from atmospheric contamination, arc stabilizers etc.

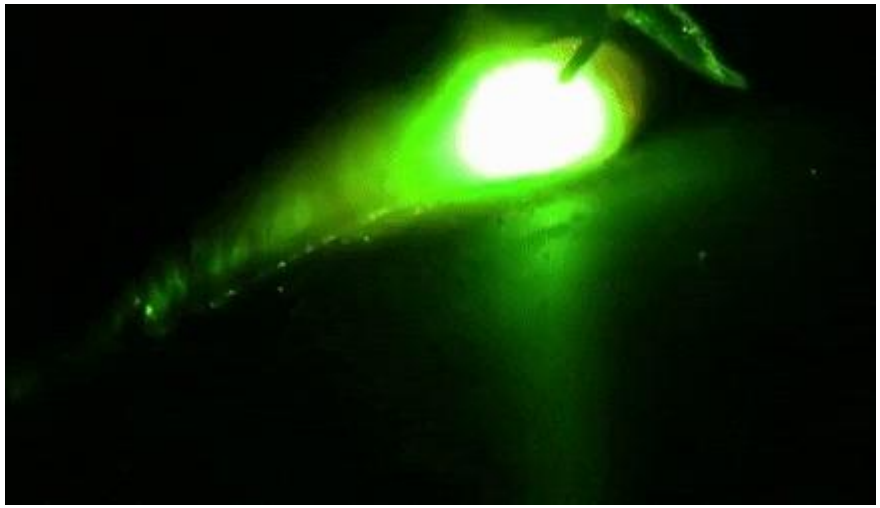
Non-consumable electrodes are made up of high melting point materials like carbon, pure tungsten or alloy tungsten etc. These electrodes do not melt away during welding. But practically, the electrode length goes on decreasing with the passage of time, because of oxidation and vaporization of the electrode material during welding. The materials of non-consumable electrodes are usually copper coated carbon or graphite, pure tungsten, thoriated or zirconiated tungsten.





electric arc completes the circuit.

Set-Up for Manual Metal Arc Welding



Metal Inert Gas (M.I.G) Welding

- Continuous wire welding in a shielded atmosphere is often identified by the abbreviations M.I.G. (Metal Inert Gas) and M.A.G. (Metal Active Gas) or G.M.A.W. (Gas Metal Arc Welding).

- Continuous wire welding is a process in which the heat required to carry out the weld is supplied by an electric arc that is maintained between the piece to be welded and the wire electrode.

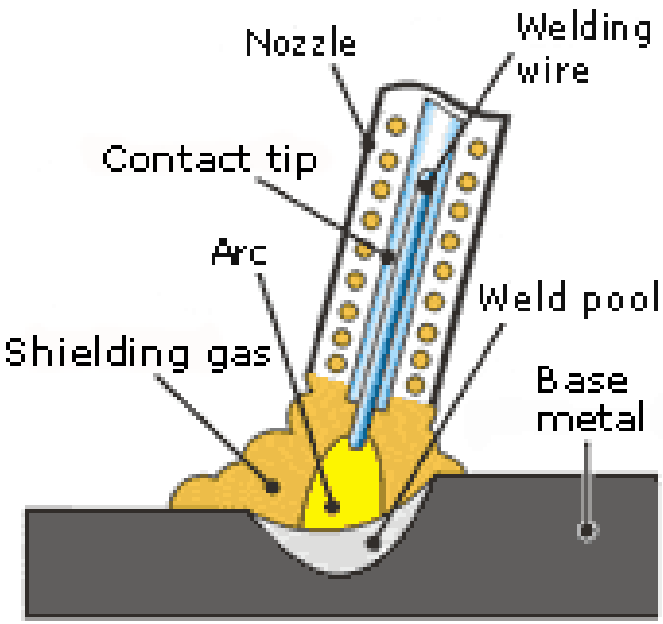
- The welding zone is constantly fed with the welding material (the wire electrode) by means of a special torch, which also supplies the flow of gas (or gas mixture) whose purpose is to shield the wire electrode, the weld pool, the arc and the area surrounding the base material from atmospheric contamination.

- The presence in the welding circuit of the gas cylinder (inert or active gas, or a mixture of the two) together with the use of solid wire electrodes, identifies the gas shield welding process (M.I.G. or M.A.G.).

- The absence of the gas cylinder from the welding circuit, together with the use of tubular wire electrodes, identifies the welding process without gas shield (SELF SHIELDED WIRE, NO GAS or FLUX); in this case the gas shield is obtained by the action of the flux core contained in the wire.

The welding circuit consists essentially of the following elements:

1. Power source
2. Torch with cable bundle
3. Wire feeder
4. Water cooling unit
5. Gas cylinder with regulation system
6. Clamp with earth cable



Power source

The purpose of the power source is to feed the welding zone with the welding material, by means of a special torch, and to maintain the electric arc that is struck between the piece to be welded and the consumable wire electrode.
Unlike M.M.A. and T.I.G power sources, where there is only one regulating parameter (welding current), on M.I.G. power sources there are two regulating devices, one which regulates electric arc intensity (welding voltage), and another which regulates the rate of welding wire feed (welding current).

Direct current (DC) power sources

Direct current power sources are the commonest power sources and are characterized by their high stability; this is because they are based on the fact that an electric arc will tend to stabilize naturally if it is powered at constant voltage and generated on a wire fed at a constant rate. Given the flexibility of the process, there is some flexibility in the choice of voltage and wire feed rate parameters. In this way it is possible to obtain drop transfer, from the welding material to the material to be welded, using either the "short arc" immersion procedure or the "spray arc" procedure.

THE SHIELD GASES

The shield gases used in M.I.G.-M.A.G. welding procedures can be divided into two basic categories: inert and active. Argon, helium and argon-helium mixtures belong to the first type, while carbon dioxide and argon/oxygen or argon/carbon dioxide mixtures are considered active gases.

Argon (Ar) is an inert gas, produced by fractional distillation of the atmosphere. The gas is extracted from the air and may, therefore, contain traces of impurities such as oxygen, nitrogen or water vapour, but it is nevertheless considered suitable for most welding applications. The use of this gas in M.A.G. applications gives good arc stability and an easy strike. Moreover, given its low thermal conductivity, the central part of the arc column remains at a high temperature giving more fluidity to the drops of material passing through the arc zone.

Helium (He) is an inert, rather rare gas that is scarcely present in the atmosphere and is extracted from underground: it is therefore much more costly than argon. Compared to argon, helium has a less stable arc but greater penetration; it is mostly used for thick welds and for materials with high thermal conductivity, such as, for example, copper and aluminium. Since helium, unlike argon, is lighter than air and hence more volatile, a greater quantity of gas is needed to ensure a sufficient shield for the welding zone.

Carbon dioxide (CO_2) is an active gas, present in the air and underground. The commonest problem caused by this type of shield is that it can cause the formation of excessive spray and establish an unstable arc; if the arc is kept rather short and of constant length, however, it is possible to keep it under control. With a CO_2 shield good penetration is generally obtained.

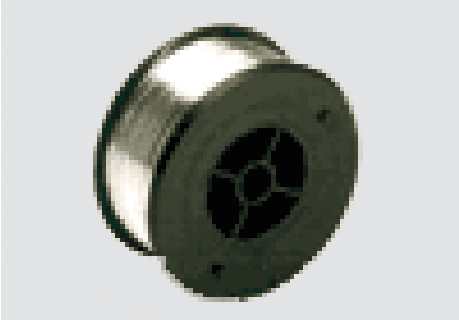
WELDING WIRES

The wires can be identified by their chemical composition and also their section morphology, which may either consist of a single metal (solid wires) or have an internal core containing granules (tubular wire). Particular attention should be paid to the presence of grease or moisture on the surface of the wire electrode, because they could cause cracks, porosity or blowholes; in addition, if the wire electrode reel is not wound uniformly this could cause uneven wire feed resulting in unstable welding.

Solid wires usually have the same composition as the base material, with added elements able to help clean the base material. The most widely used diameters are 0.6 – 0.8 – 0.9 - 1 – 1.2 – 1.6 mm.

Tubular wires, with gas shield, do not consist of solid metal but have an internal core filled with granular powder (flux); this has the same functions as the coating on coated electrodes.

The granular powder or flux can be of rutile, basic or special type. Compared to solid wires, tubular wires have better arc stability and deeper penetration, ensure a better-looking seam, often eliminating the need for further finishing (e.g. spray grinding) and reducing the risk of defect formation, such as porosity; of course the use of tubular electrodes requires slag removal, as for welding with M.M.A. electrodes. The most widely used diameters are 0.6 – 0.8 – 0.9 – 1.2 – 1.6 mm.



MATERIALS WELDED BY M.I.G.

1. Mild, carbon steels

Carbon steels are welded with direct current and electrode positive polarity (the wire electrode is connected to the positive pole) using exclusively the M.I.G. welding procedure. Applications may range from the use of CO_2 only, to argon - CO_2 mixtures in differing proportions (the most common is 80% argon, 20% CO_2). The greater the percentage of argon in the mixture the better the properties and arc stability. The weld has very good properties, especially with short arc where dense/thick weld pools are obtained, allowing welding in all positions. The use of wires with silicon and manganese content will lead to the elimination of impurities present in the base material and obtain good quality welds. Bevelled seams should be prepared with thicknesses greater than 3 mm.

2. Stainless steels

Stainless steels are welded with direct current and electrode positive polarity (the wire electrode is connected to the positive pole) using exclusively the M.I.G. welding procedure. The gas shield used should consist of Ar + CO_2 or Ar + Oxygen mixtures. The proportion of argon in the mixture should never fall below 98% so as to avoid strong oxidation of the chrome present in the base material. Seams thicker than 2.5 mm should be bevelled. Care should be taken to choose the most suitable welding material for the quality of the stainless steel being welded. To achieve a good weld, we advise grinding the spots.

3. Aluminum and its alloys

Aluminum and its alloys are welded in direct current or pulsed current with electrode positive polarity (the wire electrode is connected to the positive pole) using the M.I.G. welding procedure. The gas shield used is usually pure argon. Pure helium or an argon + helium mixture can also be used. For horizontal welding, irrespective of the thickness, the spray-arc and/or pulsed-arc technique is used; the short-arc technique is used, instead, on thin vertical or corner welds. Given the properties of aluminum, milling the spots instead of grinding is advisable.

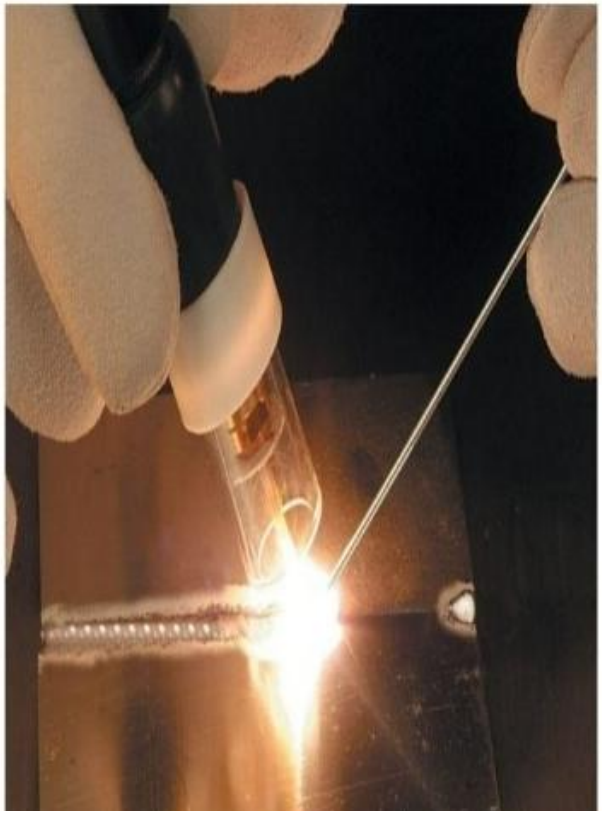
4. Other materials

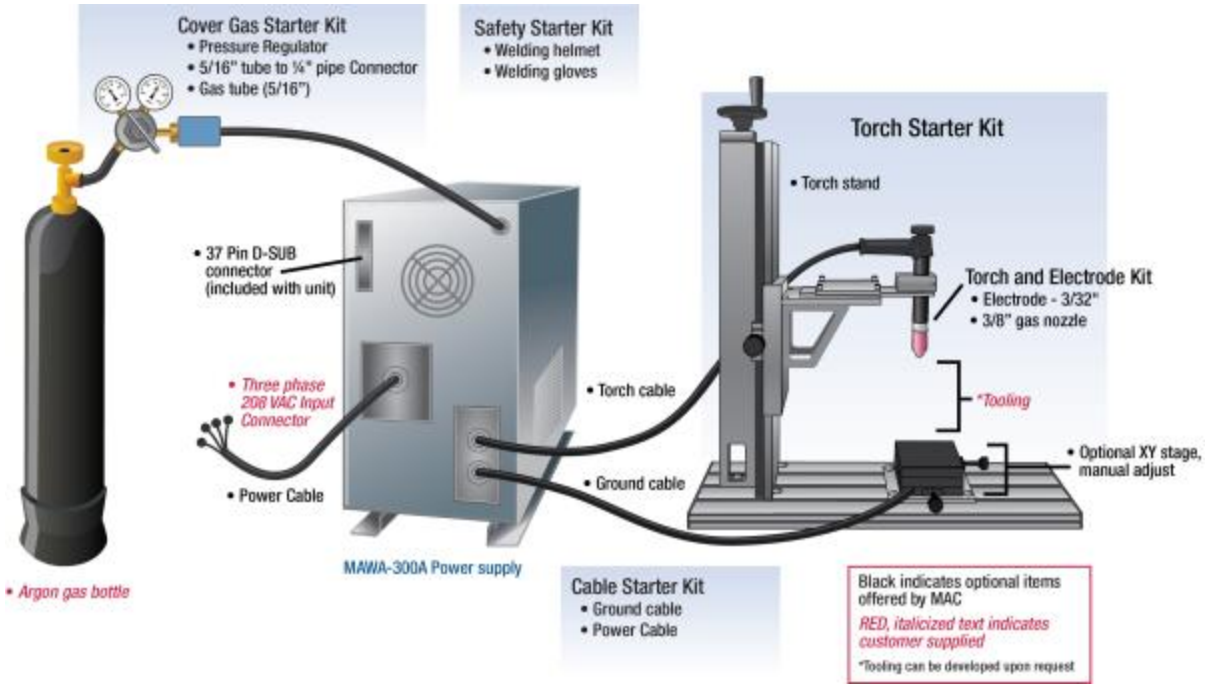
The M.I.G. welding procedure is also used on materials such as nickel and its alloys, copper and its alloys; for all these materials direct current with electrode positive polarity is used. For welding copper more than 5 mm thick we advise adoption of the M.I.G. procedure but in any case the current intensity should be adjusted according to the welding position and the seam thickness.

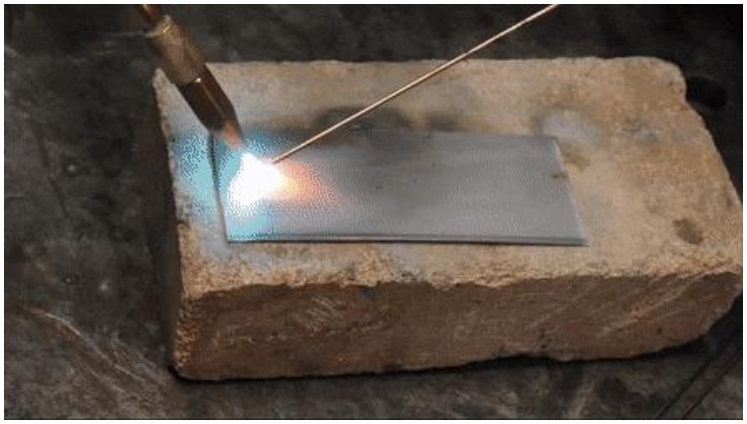
TIG WELDING

- *Background*

- Initially developed in 1941 by Russel Meredith for welding magnesium, as its oxides have high m.p. than parent metal
- TIG designated nowadays as GTAW(Gas tungsten arc welding)
- Currently used for welding aluminium, stainless steel, titanium, nickel, copper, etc.
- Highly attractive replacement for gas welding as inert gas is used for shielding instead of slag




Cover Gas Starter Kit
• Pressure Regulator
• 5/16" tube to ¼" pipe Connector
• Gas tube (5/16")
Safety Starter Kit
• Welding helmet
• Welding gloves
Torch Starter Kit
• Torch stand
Torch and Electrode Kit
• Electrode - 3/32"
• 3/8" gas nozzle
• 37 Pin D-SUB connector (included with unit)
• Three phase 208 VAC Input Connector
• Power Cable
• Torch cable
• Ground cable
*Tooling
• Optional XY stage, manual adjust
• Argon gas bottle
MAWA-300A Power supply
Cable Starter Kit
• Ground cable
• Power Cable
Black indicates optional items offered by MAC
RED, italicized text indicates customer supplied
*Tooling can be developed upon request



Tungsten Inert Gas (TIG)

- TIG stands for tungsten inert gas welding or sometimes this welding is known as gas tungsten arc welding.

- In this welding process, the heat required to form weld is provided by a very intense electric arc which is form between tungsten electrode and work piece.

- In this welding a non-consumable electrode is used which does not melt.

- Mostly no filler material is required in this type of welding but if it required, a welding rod fed into the weld zone directly and melted with base metal.

- This welding is mostly used for welding aluminum alloy.

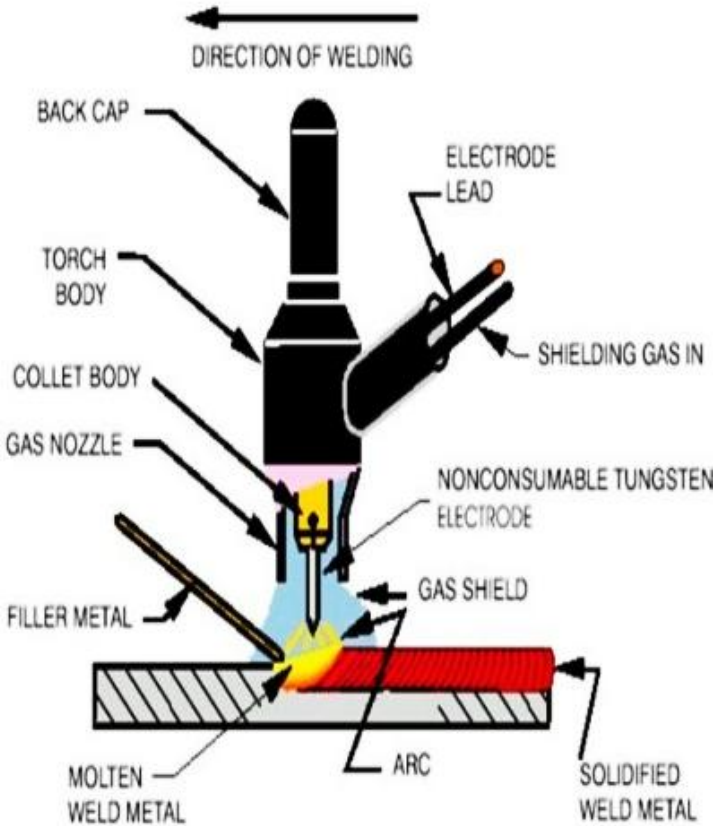
Principle

- TIG welding works on same principle of arc welding.
- In a TIG welding process, a high intense arc is produced between tungsten electrode and work piece.
- In this welding mostly work piece is connected to the positive terminal and electrode is connected to negative terminal.
- This arc produces heat energy which is further used to join metal plate by fusion welding.
- A shielding gas is also used which protect the weld surface from oxidization.

THE TIG PROCESS

- *Working*

- Arc is struck between the non-consumable tungsten electrode and the work piece to fuse metals
- Arc is covered by a layer of shielding gas which acts as the flux and keeps the nitrogen and oxygen in the air from coming in contact with the molten puddle.
- When the puddle is formed on the base metal, the torch is moved along the joint until the workpiece is fused together
- A filler rod may or may not be used
- If a filler rod is used, it should be the same composition as the base metal.
- The filler rod is fed manually into the leading edge of the puddle.
- The torch may be moved in a semicircular motion to vary the width of the bead.



FEATURES OF TIG WELDING

- Temperature may be in the range of 10000k
- Automatic and manual techniques can be used
- TIG may be done in all positions.
- Metal thickness upto 5mm
- Weld speed from 8mm/s to 50mm/s
- With superior arc and weld puddle control, clean welds are produced
- No sparks or spatter because only the necessary amount of filler metal is added to the welding pool.
- No smoke and fumes unless
- No flux required as argon gas protects the weld pool from contamination

TIG WELDING EQUIPMENT

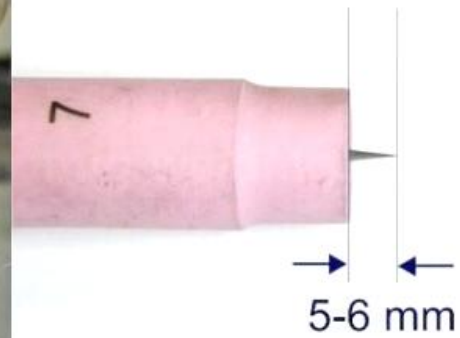
- Welding current generator DCSP/AC
- Electrode lead hose for current, may be a woven tube or flexible cable & shielding gas hose
- Gas cylinder with flowmeter
- Welding torch

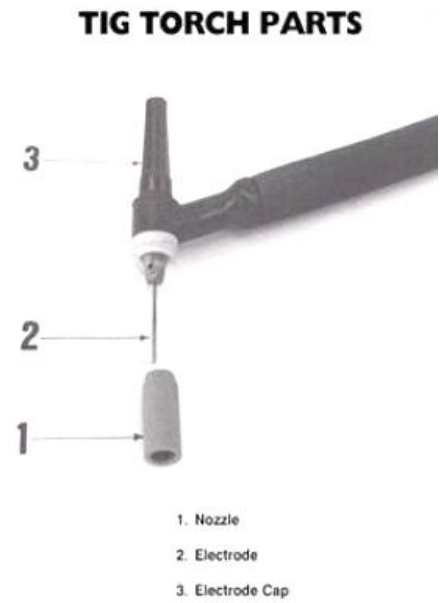
FLOWMETER

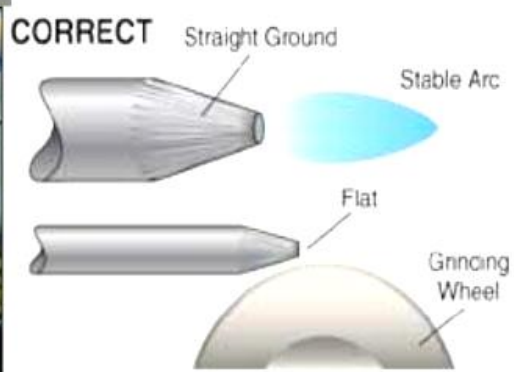
Flowmeter and tank gauge for argon and helium shielding gases.

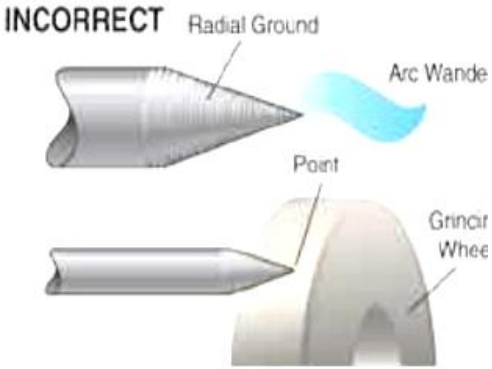
TIG TORCH PARTS, ORIENTATION & ELECTRODE GRINDING

- Nozzle for gas flow
- Collet to hold tungsten electrode
- Gas cap made of ceramic
- Tungsten electrode to be grinded before inserting into collet









TIG SHIELDING GASES
ARGON VS HELIUM

Argon

- Heavier in weight
- Good arc starting
- Less base metal distortion
- Good cleaning action
- Good arc stability
- Focused arc cone
- Lower arc voltages
- 10-30 CFH flow rates
- For thin sheets

Helium

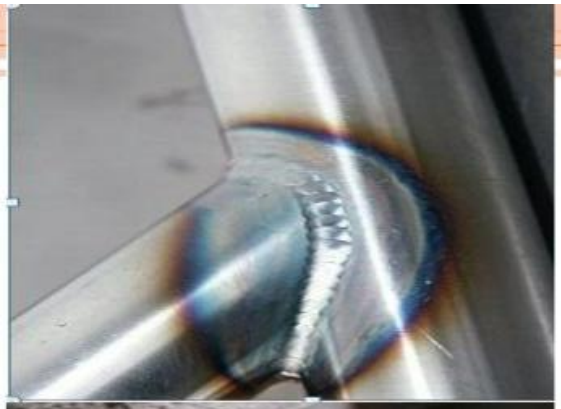
- Faster travel speeds
- Increased penetration
- Difficult arc starting
- Less cleaning action
- Flared arc cone
- Higher arc voltages
- Higher flow rates (2x)
- Higher cost than argon
- For thick sheets

ELECTRODE USAGE

- Pure tungsten electrodes have low current capacity
- To improve the electrical conductivity, small amounts of thoria or zirconia
- Electrodes with 1 per cent thoria are used for welds on aircraft, missiles, nuclear reactors, and heat exchangers
- Zirconiated Tungsten is most commonly used for AC welding of aluminum and magnesium alloys, preferred when Tungsten contamination of weld is intolerable.

NOZZLES

- Nozzles are made from ceramic, metal, plastic, and Pyrex glass materials
- Ceramic nozzles are used on jobs up to 275 amps.
- Metal nozzles or metal-coated ceramic nozzles are used on jobs where 300 or more amps of current are needed.



Pure

2% Ceriated

1.5% Lanthanated

.8% Zirconiated

2% Thoriated

TIG EQUIPMENT SPECIFICATIONS

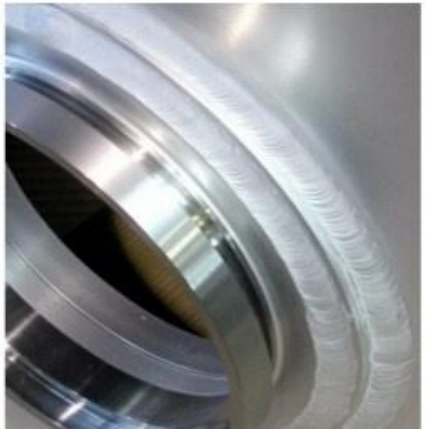
TIG WELDING MACHINE Technical Specifications

PARAMETERS	INVTIG-200	INVTIG-250	INVTIG-400	INVTIG-500
Rated Input Supply Voltage	220 V AC	415 V AC	415 V AC	415 V AC
Phases	1	3	3	3
Frequency	50	50	50	50
Max Rated Input Power in KVA	10	10	19	25
Open Circuit Voltage	55 V	65 V	65 V	65 V
Rated duty cycle	35.00%	60.00%	60.00%	60.00%
Insulation Class	IP 21	IP 21	IP 21	IP 21
Cooling	Forced Air	Forced Air	Forced Air	Forced Air
Welding Current Range	10 – 200 A	10 – 250 A	5 – 400 A	5 –500 A
Dimension in mm	395X153X301	505X203X375	494X327X360	650X310X640
Weight in Kgs	9	19	28	42

ADVANTAGES & DISADVANTAGES

- *Advantages*
- Welds more metals & metal alloys
- High quality and precision
- Pin point control
- Aesthetic weld beads
- No sparks or spatter
- No flux or slag
- No smoke or fumes

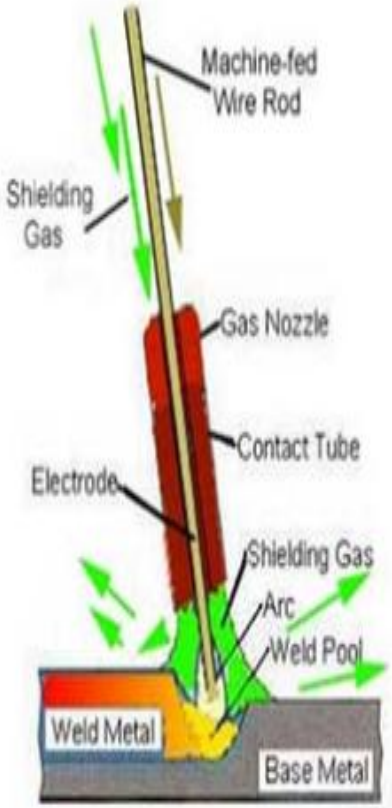
- *Disadvantages*
- Lower filler metal deposition rates
- Good hand-eye coordination a required skill
- Brighter UV rays than other processes
- Slower travel speed than other processes
- Equipment costs tend to be higher



Applications

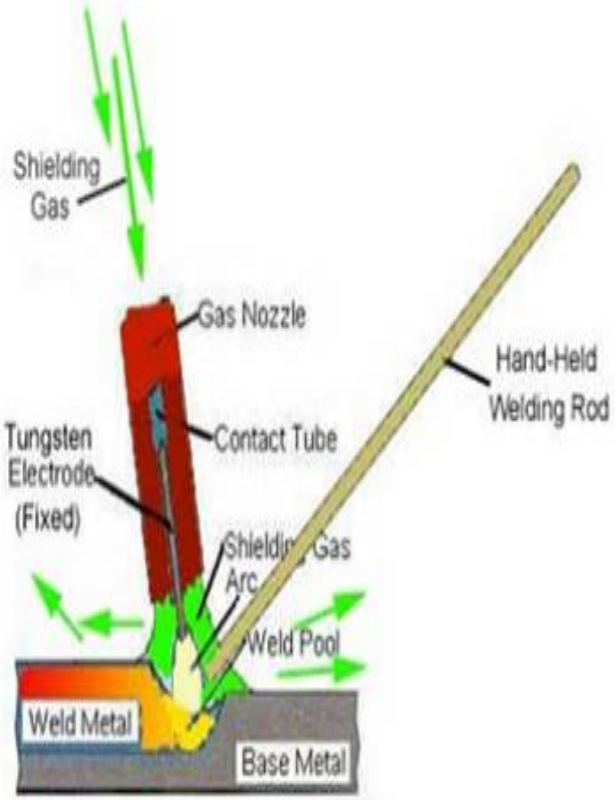
- Mostly used to weld aluminum and aluminum alloys.
- It is used to weld stainless steel, carbon base alloy, copper base alloy, nickel base alloy etc.
- It is used to welding dissimilar metals.
- It is mostly used in aerospace industries.

MIG WELDING	TIG WELDING
1. This welding is known as metal inert gas welding.	1. This is known as tungsten inert gas welding.
2. Metal rod is used as electrode and work piece used as another electrode.	2. Tungsten rod is used as electrode.
3. It is gas shielded metal arc welding.	3. It is gas shielded tungsten arc welding.
4. Continuous feed electrode wire is used which are fast feeding.	4. Welding rods are used which are slow feeding.
5. The welding area is flooded with a gas which will not combine with the metal.	5. Gas is used to protect the welded area form atmosphere.
6. MIG can weld materials such as mild steel, stainless steel and aluminum. A range of material thicknesses can be welded from thin gauge sheet metal right up to heavier structural plates.	6. TIG weld things like kitchen sinks and tool boxes. Pipe welding and other heavier tasks can also be performed, you just need to have a unit that is capable of putting out the amount of power that you need.
7. MIG requires consumable metallic electrode.	7. It used non consumable tungsten electrode
8. Electrode is feeded continuously from a wire reel.	8. It does not require electrode feed.
9. DC with reverse polarity is used.	9. It can use both A.C and D.C.
10. Filler metal is compulsory used.	10. Filler metal may or may not be used.
11. It can weld up to 40 mm thick metal sheet.	11. Metal thickness is limited about 5 mm.
12. MIG is comparatively faster than TIG.	12. TIG is a slow welding process.



MIG

Metal Inert-Gas



TIG

Tungsten Inert-Gas

Thank You